Assignment 7

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Title
Development of a Student Feedback Review Web Application using React.js, Express.js, and MongoDB

Objectives
The main objectives of this assignment are:

1. To build a complete full-stack application using the MERN stack.
2. To design a system for collecting and managing student feedback.
3. To understand how frontend and backend communicate using APIs.
4. To implement features like feedback submission, filtering, and visualization.
5. To gain practical experience in developing interactive web applications.

Problem Statement
The aim of this project is to develop a Student Feedback Review System using React.js for frontend, Express.js for backend, and MongoDB for database management.
The application should allow users to:

- Submit feedback related to courses and instructors
- View all submitted feedback
- Apply filters and sorting options
- Analyze feedback using statistical insights

The system should provide a user-friendly interface and efficiently store feedback data in MongoDB.

Outcomes
After completing this assignment, the following outcomes are achieved:

1. Ability to create dynamic UI using React.js components.
2. Development of backend APIs using Express.js.
3. Integration of frontend and backend using REST APIs.
4. Efficient data handling using MongoDB.
5. Understanding of full MERN stack application development.

Hardware and Software Requirements
Hardware Requirements

- Computer / Laptop
- Minimum 4GB RAM

Software Requirements

- Node.js
- MongoDB
- React.js
- Visual Studio Code
- Web Browser (Chrome / Firefox)

Theory

1. React.js
React.js is a frontend JavaScript library used to create interactive user interfaces. It divides the UI into reusable components and uses a virtual DOM to improve performance.

2. Node.js
Node.js is a runtime environment that allows server-side execution of JavaScript. It is used to handle requests and manage communication with the database.

3. Express.js
Express.js is a backend framework that simplifies the development of APIs and web applications by handling routing and middleware.

4. MongoDB
MongoDB is a NoSQL database that stores data in a flexible document format. It is ideal for applications where data can vary, such as feedback systems.

5. MERN Stack
The MERN stack consists of:

- MongoDB (Database)
- Express.js (Backend framework)
- React.js (Frontend library)
- Node.js (Runtime environment)

This stack enables smooth communication between frontend and backend.

6. Feedback Review System
The system is designed to collect and display feedback effectively. It includes:

- A form for submitting feedback
- A section for viewing reviews
- Filtering and sorting options
- An analytics section for insights

7. Features of the Application
1. Feedback Submission
Users can provide feedback by entering:

- Student name
- Course name
- Instructor name
- Category (e.g., Teaching, Content)
- Rating
- Comments

3. Filtering and Sorting

Users can:

- Filter feedback by course or instructor
- Sort feedback based on rating or date

This improves usability and helps in analysis.

4. Analytics Dashboard

The system provides insights such as:

- Course-wise performance
- Category-wise feedback
- Distribution of ratings

These insights help in evaluating teaching quality.

8. Working of the Application

The system works as follows:

1. The user opens the web application.
2. The frontend interface loads using React.
3. The user navigates to the feedback form.
4. Feedback is entered and submitted.
5. Data is sent to the backend via API.
6. The backend processes and stores the data in MongoDB.
7. Updated data is fetched from the database.
8. Feedback and analytics are displayed on the interface.

Flowchart

Start
↓
User opens application
↓
Display interface
↓
Submit feedback
↓
Send data to backend
↓
Store in MongoDB
↓
Fetch updated data
↓
Display feedback
↓
Show analytics
↓
End

Design / Execution Steps

The following steps were followed:

1. Installed Node.js, MongoDB, and React.js.
2. Created a React frontend application.
3. Designed UI components for feedback system.
4. Developed backend server using Express.js.
5. Connected MongoDB database.
6. Created schema for storing feedback.
7. Developed API routes for handling requests.
8. Integrated frontend with backend using APIs.
9. Implemented filtering and sorting functionality.
10. Added analytics dashboard for visualization.
11. Tested the application thoroughly.

Test Cases:

Test Case	Description	Expected Result
Application Load	Open website	UI loads properly
Submit Feedback	Enter details	Data stored successfully
View Reviews	Open reviews page	Feedback displayed
Apply Filters	Filter data	Correct results shown
Analytics View	Open dashboard	Data visualized correctly

Code:

HTML: Dashboard Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Insight Spark</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- Header -->
  <header class="header">
    <h1>InsightSpark</h1>
    <nav>
```

```
<button>Reviews</button>
<button>Submit</button>
<button>Analytics</button>
</nav>
</header>

<!-- Stats -->
<section class="stats">
<div class="card">
<h3>Total Reviews</h3>
<p>120</p>
</div>
<div class="card">
<h3>Average Rating</h3>
<p>4.2</p>
</div>
<div class="card">
<h3>Active Feedback</h3>
<p>35</p>
</div>
</section>

<!-- Feedback List -->
<section class="feedback-list">
<div class="feedback-card">
<h4>Data Structures</h4>
<p>Instructor: Dr. Sharma</p>
<p>              </p>
<p>Great teaching but needs more practical examples.</p>
</div>
</section>

</body>
</html>
```

CSS

```
body {
  font-family: 'Inter', sans-serif;
  margin: 0;
  background: #f4f6f9;
}
```

```
/* Header */
.header {
 display: flex;
 justify-content: space-between;
 padding: 15px 30px;
 background: white;
 box-shadow: 0 2px 5px rgba(0,0,0,0.1);
}

.header h1 {
 color: #2b6cb0;
}

/* Stats */
.stats {
 display: flex;
 gap: 20px;
 padding: 20px;
}

.card {
 background: white;
 padding: 20px;
 flex: 1;
 border-radius: 10px;
 text-align: center;
 box-shadow: 0 4px 10px rgba(0,0,0,0.08);
}

.card p {
 font-size: 22px;
 color: #2b6cb0;
}
```

JavaScript

```
const API = "http://localhost:5000/api";

// Fetch feedback
async function loadFeedback() {
  const res = await fetch(`${API}/feedbacks`);
  const data = await res.json();
  console.log(data);
}
```

```
// Submit feedback
async function submitFeedback() {
 const feedback = {
 studentName: "Anish",
 course: "DBMS",
 instructor: "Dr. Patil",
 rating: 4,
 comment: "Good course overall"
 };

 await fetch(`${API}/feedbacks`, {
 method: "POST",
 headers: { "Content-Type": "application/json" },
 body: JSON.stringify(feedback)
 });

 alert("Feedback Submitted");
}

loadFeedback();
```

React.js

```
import React, { useState, useEffect } from "react";

function App() {
  const [feedbacks, setFeedbacks] = useState([]);

  useEffect(() => {
    fetch("http://localhost:5000/api/feedbacks")
      .then(res => res.json())
      .then(data => setFeedbacks(data.feedbacks));
  }, []);

  return (
    <div>
      <h1>Insight Spark Dashboard</h1>

      {feedbacks.map((fb) => (
        <div key={fb._id} className="card">
          <h3>{fb.course}</h3>
          <p>{fb.instructor}</p>
```

```
<p>    {fb.rating}</p>
<p>{fb.comment}</p>
</div>
))}
</div>
);
}

export default App;
```

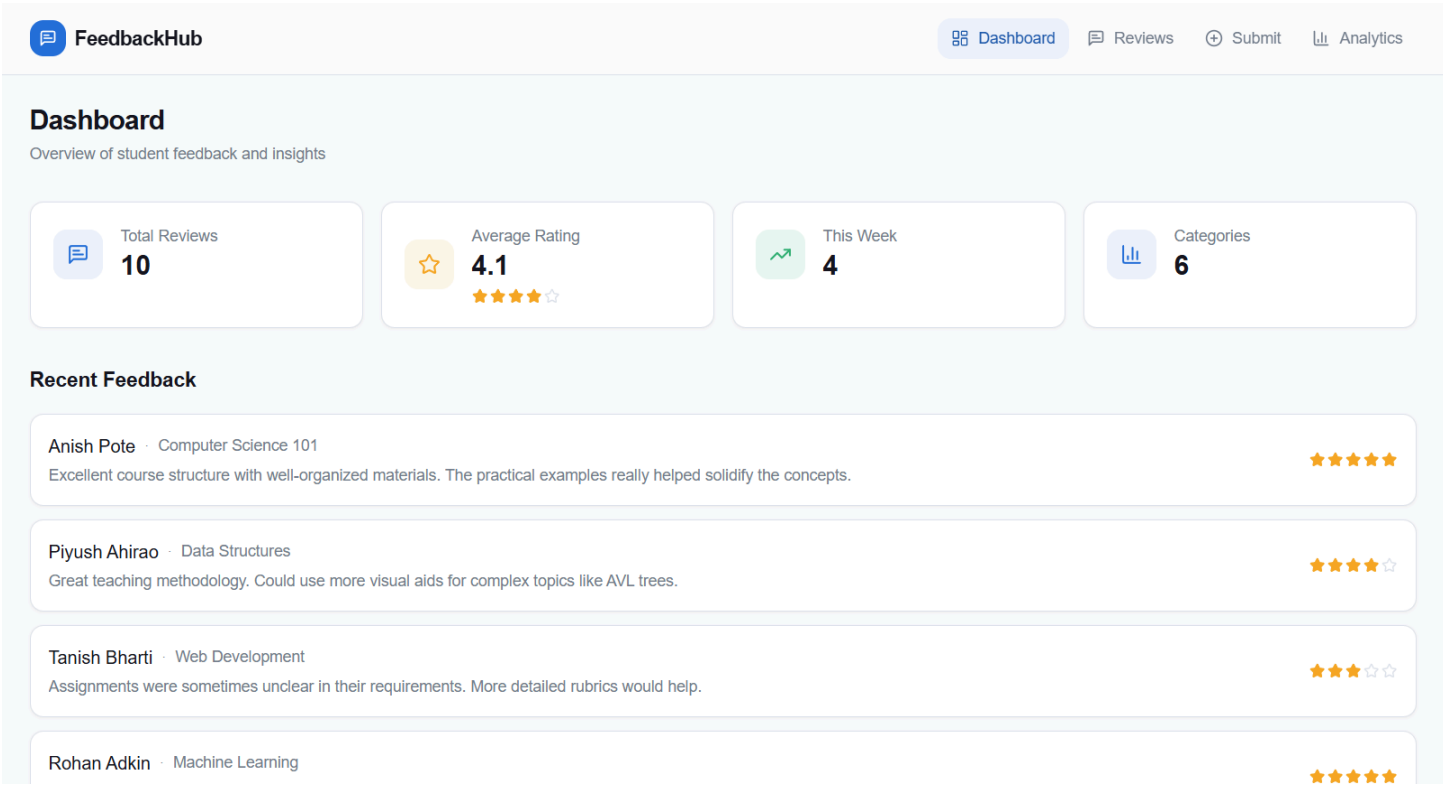
MongoDB Schema

```
const mongoose = require("mongoose");

const feedbackSchema = new mongoose.Schema({
  studentName: { type: String, required: true },
  course: { type: String, required: true },
  instructor: { type: String, required: true },
  rating: { type: Number, min: 1, max: 5 },
  comment: String,
  createdAt: { type: Date, default: Date.now }
});

module.exports = mongoose.model("Feedback", feedbackSchema);
```

Output



FeedbackHub

Dashboard | Reviews | Submit | Analytics

Reviews

10 feedback entries

Filters

Search... | All Courses | All Instructors | All Categories | Newest First

Anish Pote
Computer Science 101
★★★★★
Course Content Dr. Rajesh Sharma
Excellent course structure with well-organized materials. The practical examples really helped solidify the concepts.
Mar 20, 2026

Piyush Ahirao
Data Structures
★★★★☆
Teaching Quality Prof. Meena Iyer
Great teaching methodology. Could use more visual aids for complex topics like AVL trees.
Mar 19, 2026

Tanish Bharti
Web Development
★★★☆☆
Assessment Dr. Kavita Nair
Assignments were sometimes unclear in their requirements. More detailed rubrics would help.
Mar 18, 2026

Rohan Adkin
Machine Learning
★★★★★
Resources Prof. Arvind Desai
Outstanding supplementary materials and research paper recommendations. Top-notch course.

Chaitanya Bakare
Database Systems
★★★★☆
Support Dr. Sunita Rao
Very responsive to questions. Office hours were extremely helpful for project guidance.

Vedant Kulkarni
Software Engineering
★★★★☆
Overall Experience Dr. Rajesh Sharma
Real-world project experience was invaluable. Team collaboration skills improved significantly.

FeedbackHub

Dashboard | Reviews | Submit | Analytics

Submit Feedback

Share your experience to help us improve

Feedback Form

Student Name

Anish Pote

Course

Data Structures

Instructor

Dr. Kavita Nair

Category

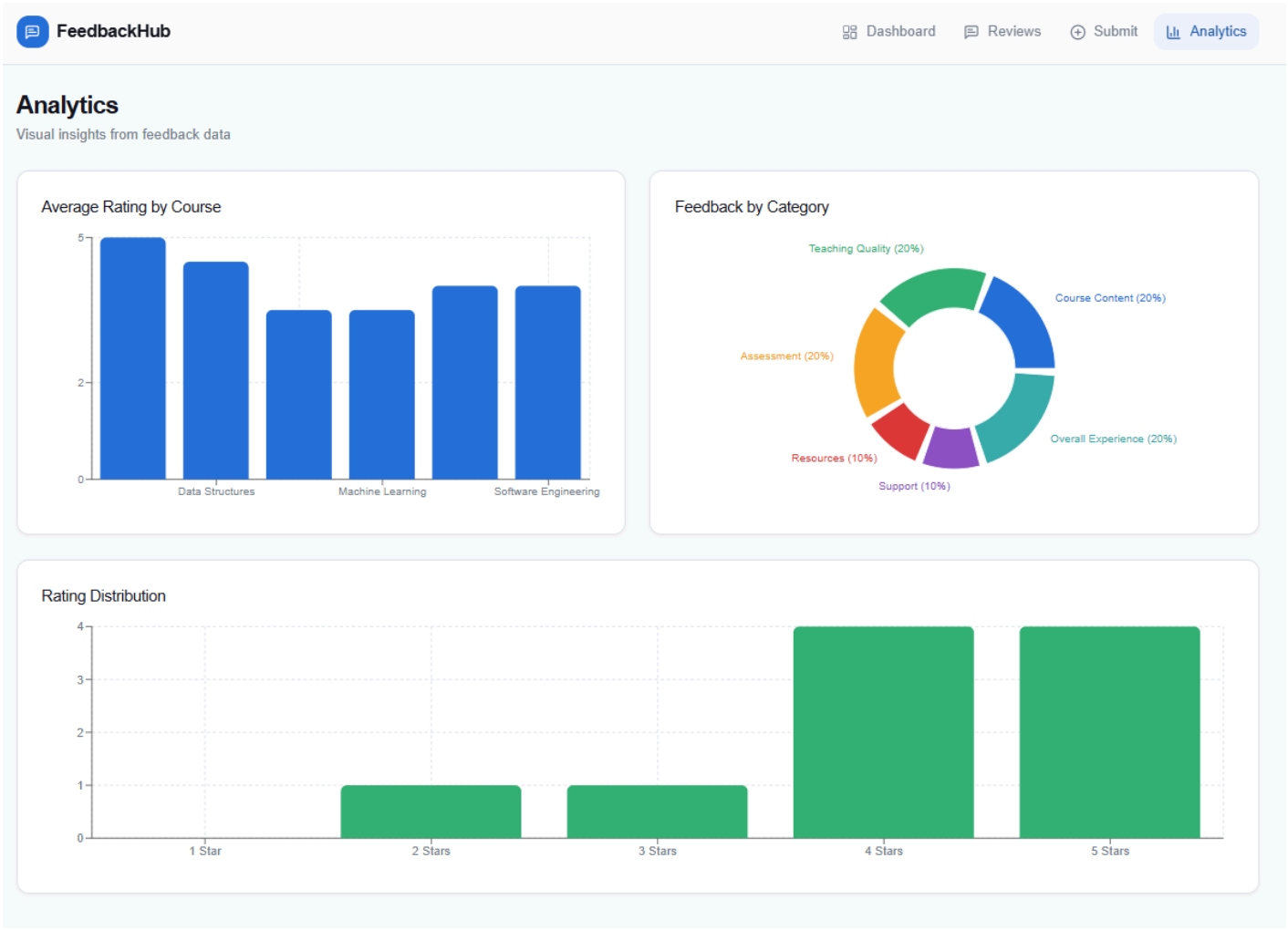
Teaching Quality

Rating

★★★★★

Comment

Share your detailed feedback...



Conclusion

In this assignment, a Student Feedback Review System was successfully developed using React.js, Express.js, and MongoDB. The application enables users to submit feedback, view reviews, and analyze data through filtering and analytics features.

This project demonstrates the practical implementation of the MERN stack and highlights how modern web applications are built with interactive interfaces and efficient backend systems.